

IMU Gesture Glove

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<https://github.com/printjan/GestureGlove>

Gesture Recognition With IMU Data

Project Idea: Control PowerPoint by recognizing gestures using an IMU glove.

Dual-IMU hardware setup:

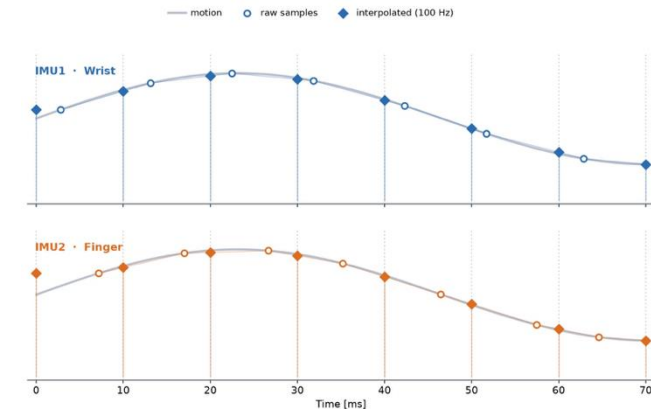
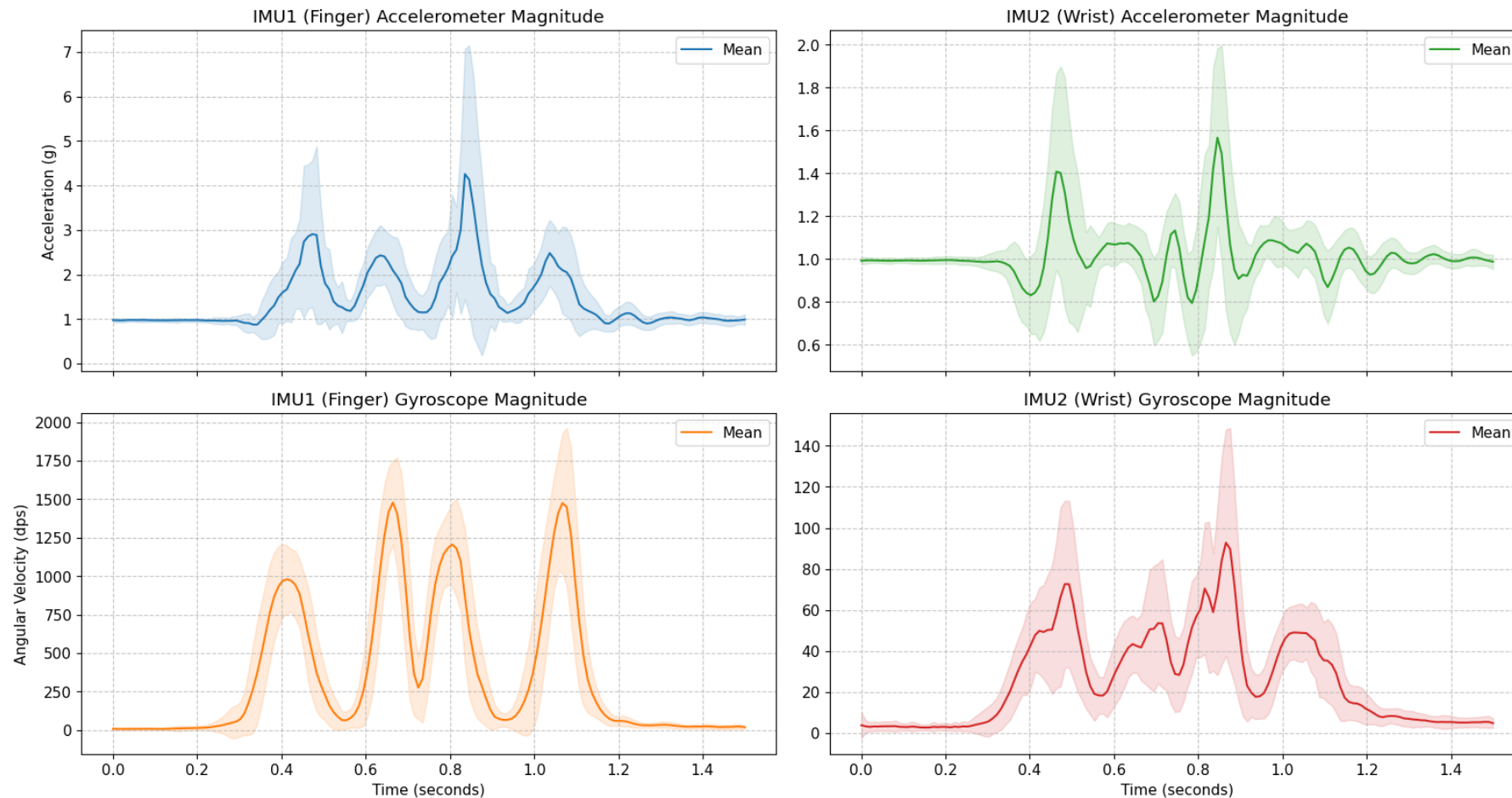
- wrist captures arm translation,
- finger captures fine-grained rotation



Gesture Recognition With IMU Data

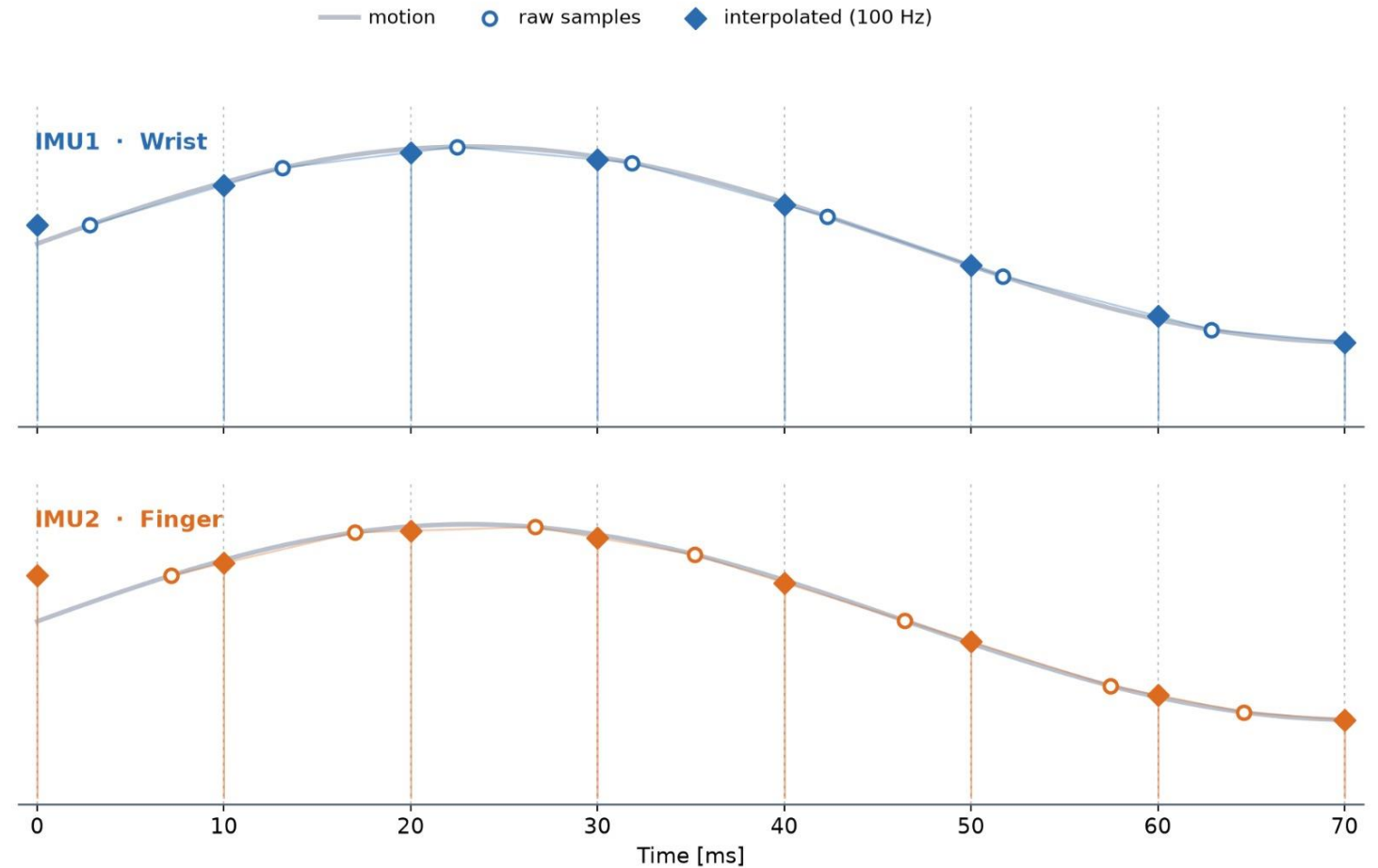
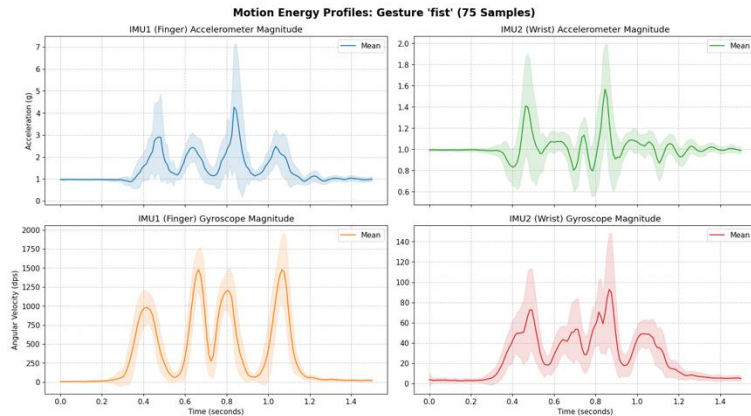
Implementation: Data Recording.

Motion Energy Profiles: Gesture 'fist' (75 Samples)



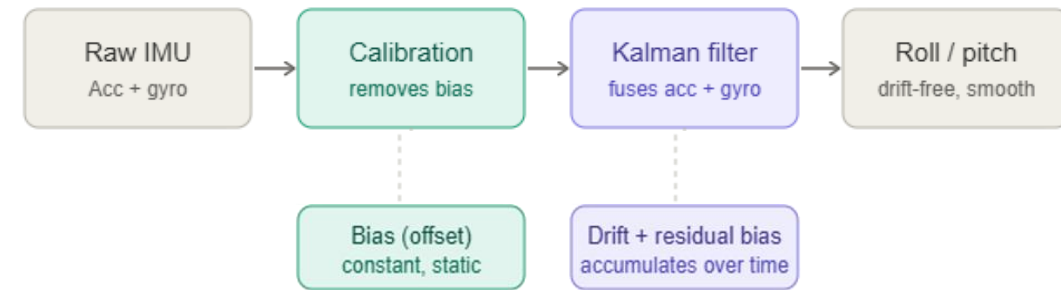
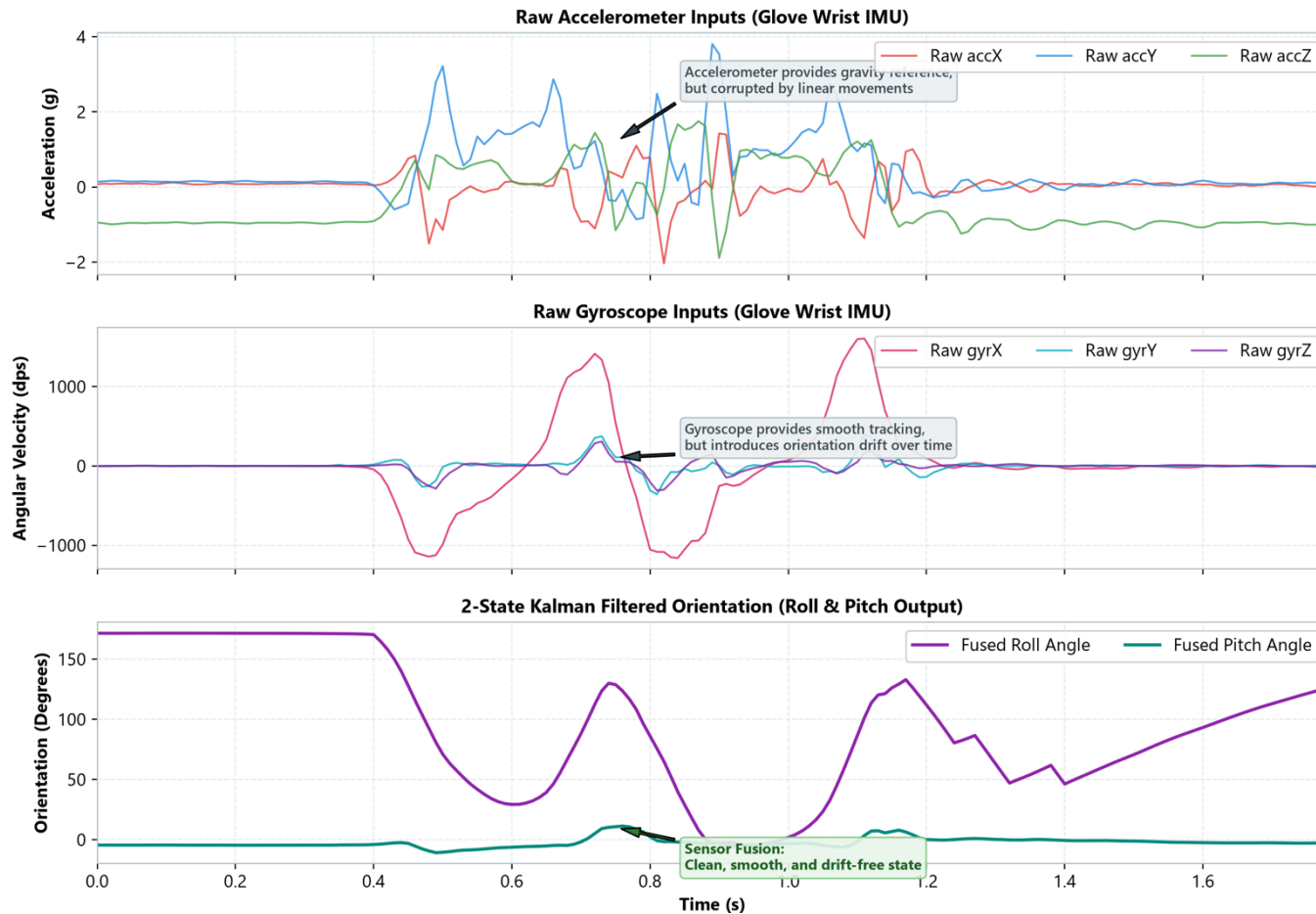
Gesture Recognition With IMU Data

Implementation: Data Recording.



Gesture Recognition With IMU Data

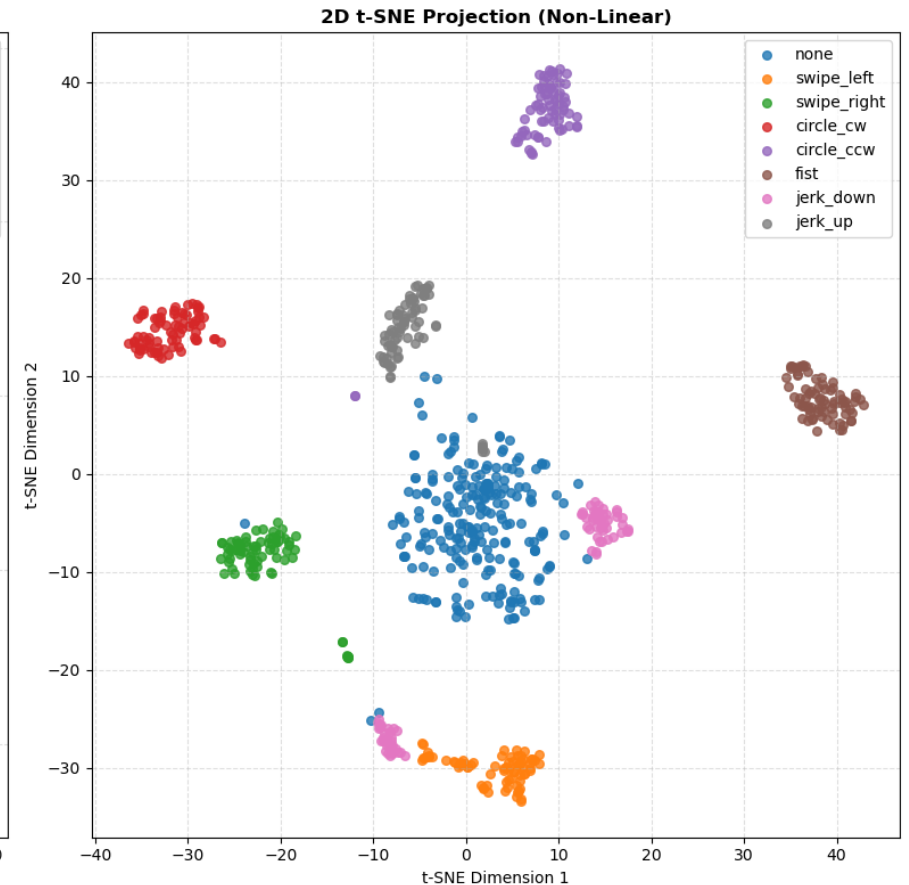
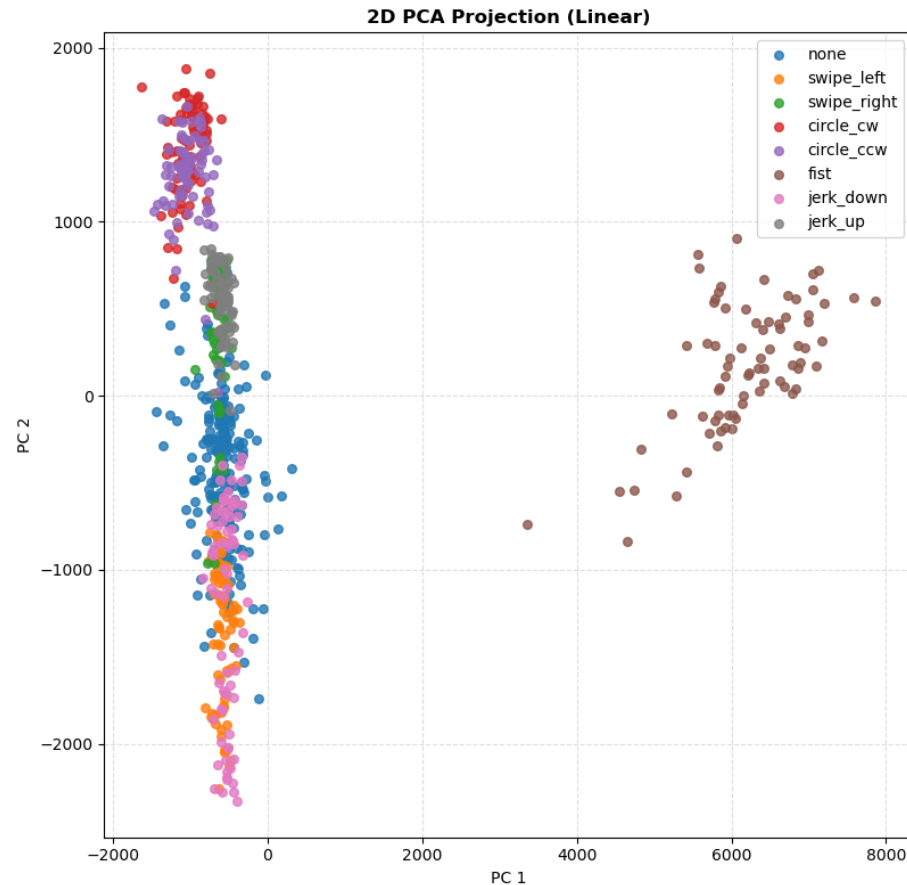
Implementation: Filtering.



Gesture Recognition With IMU Data

Implementation: Data Audit - Is this even implementable?

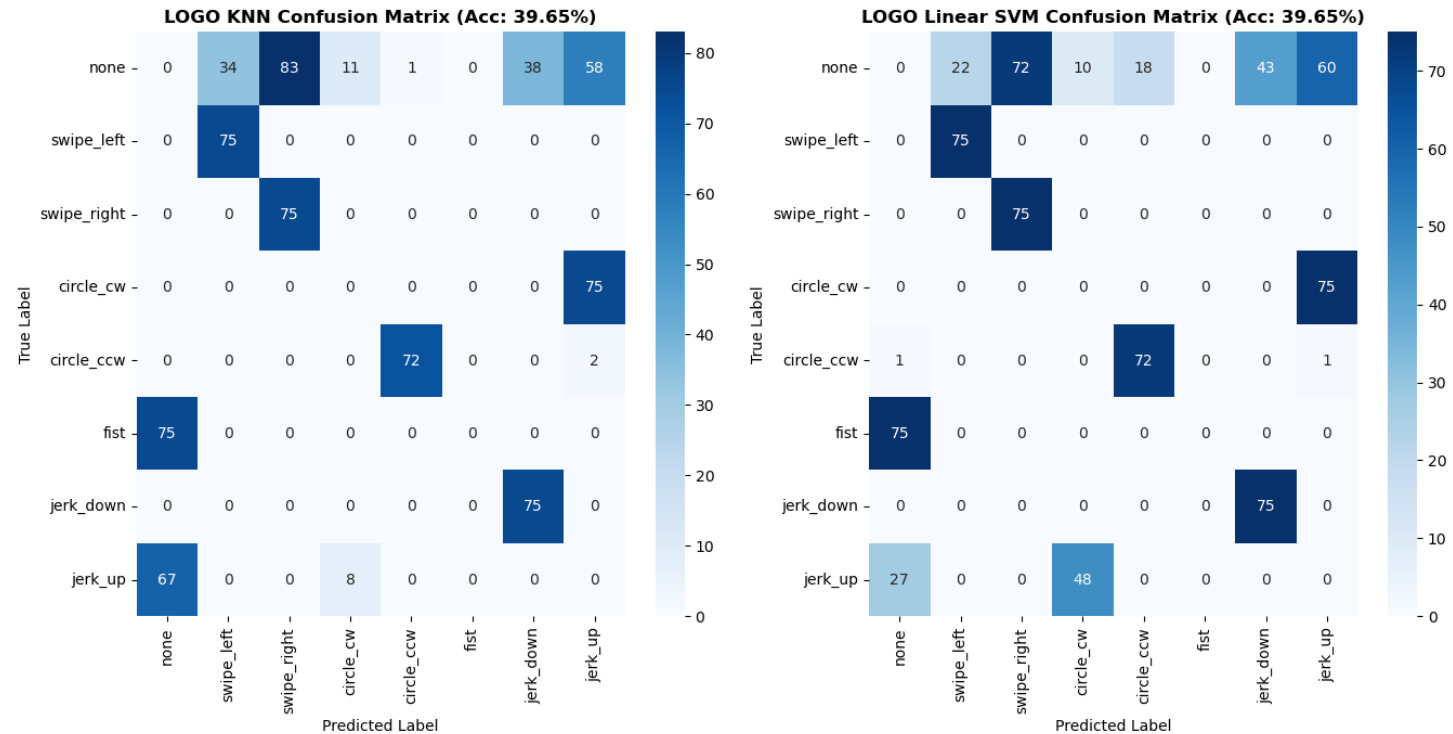
- **Linear Projection (PCA):**
no clusters at all.
- **Non-Linear Manifold (t-SNE):**
visible clustering.



Gesture Recognition With IMU Data

Implementation: Data Audit - Is this even implementable?

- KNN/SVM baseline:
only 39.6% accuracy



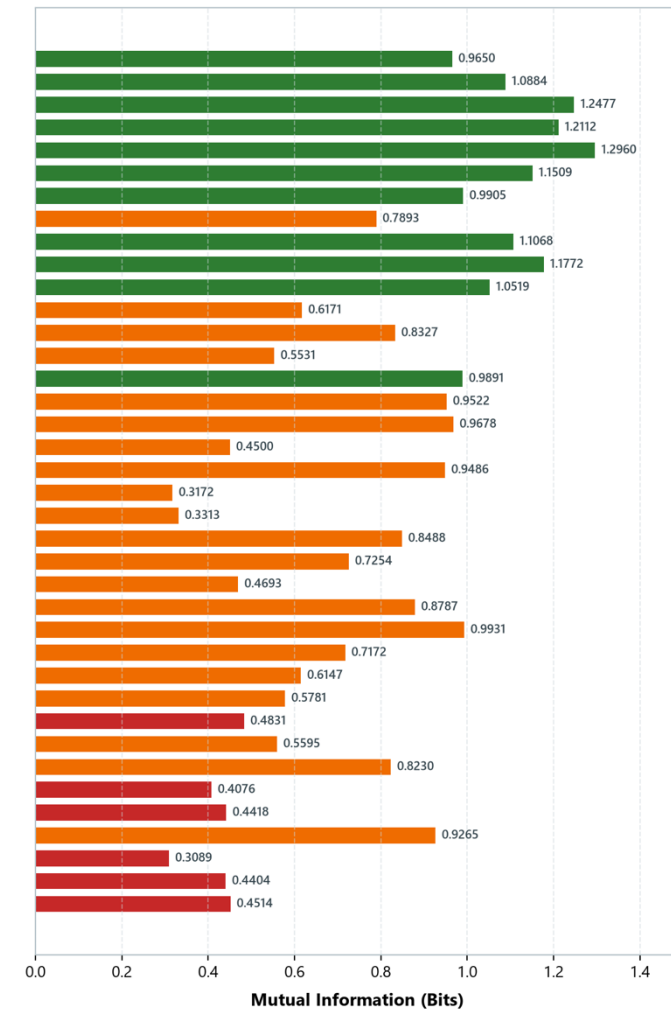
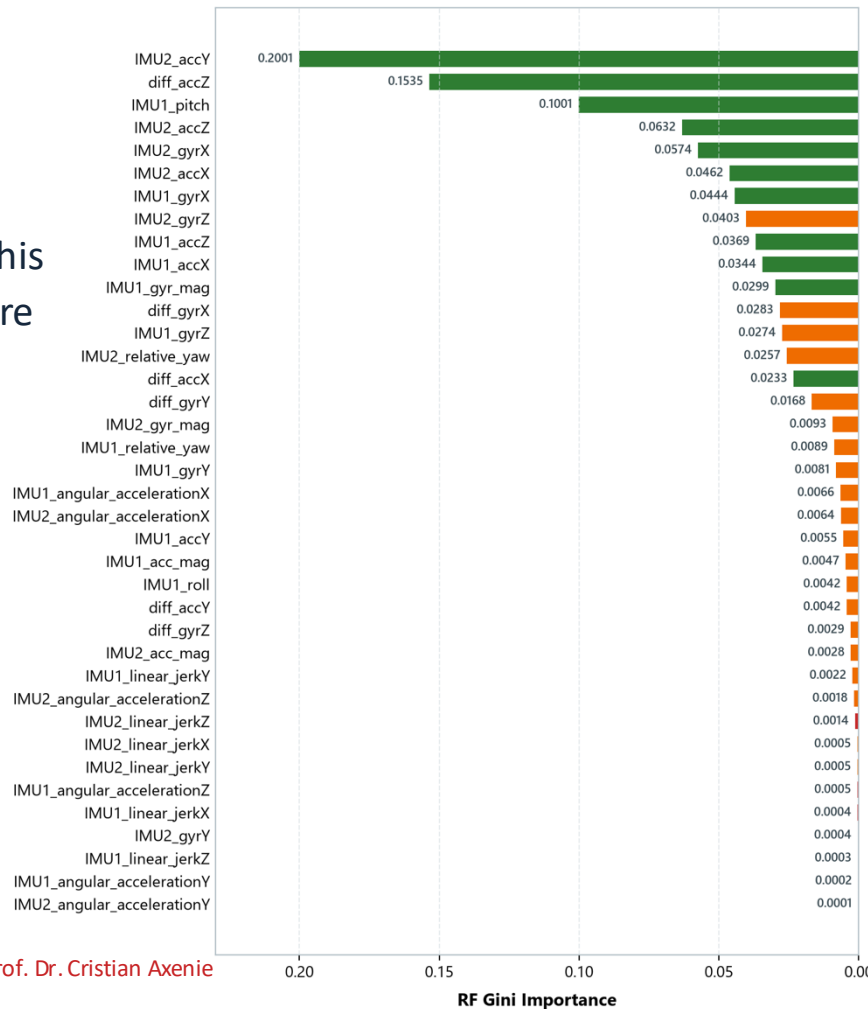
Conclusion: The data is separable, but only with models that learn temporal and spatial invariances — we need **deep learning**.

Gesture Recognition With IMU Data

Implementation: Data Audit – Feature Engineering.

Two metrics per feature:

- **Mutual Information (MI):**
how much information does this channel carry about the gesture label?
- **Random Forest Gini Importance (RF)**
how much does this channel contribute to decision boundary splits?

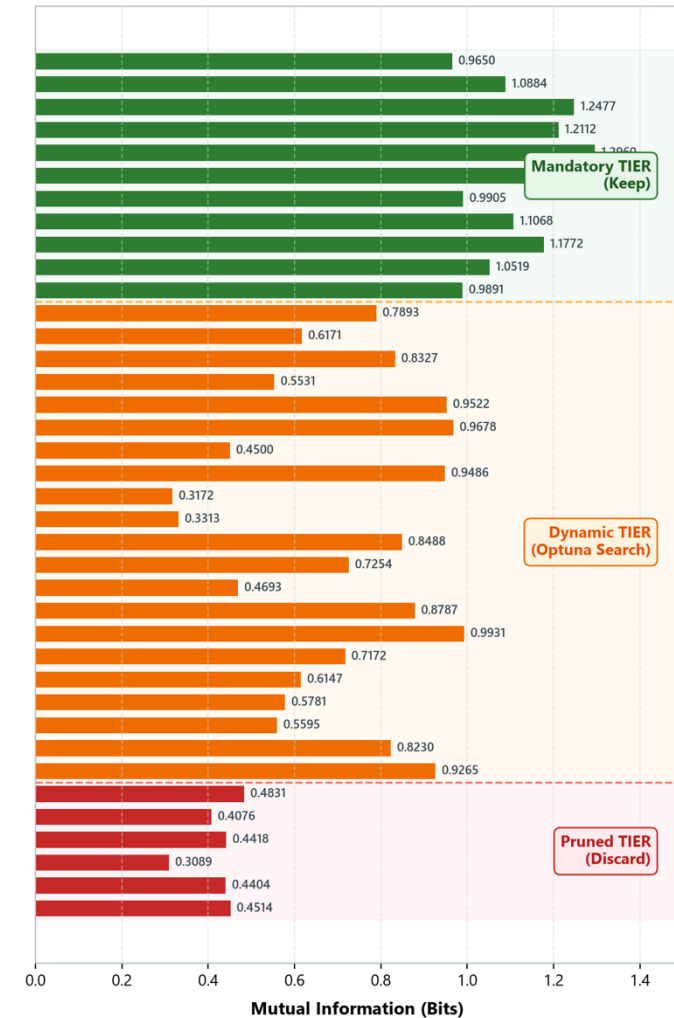
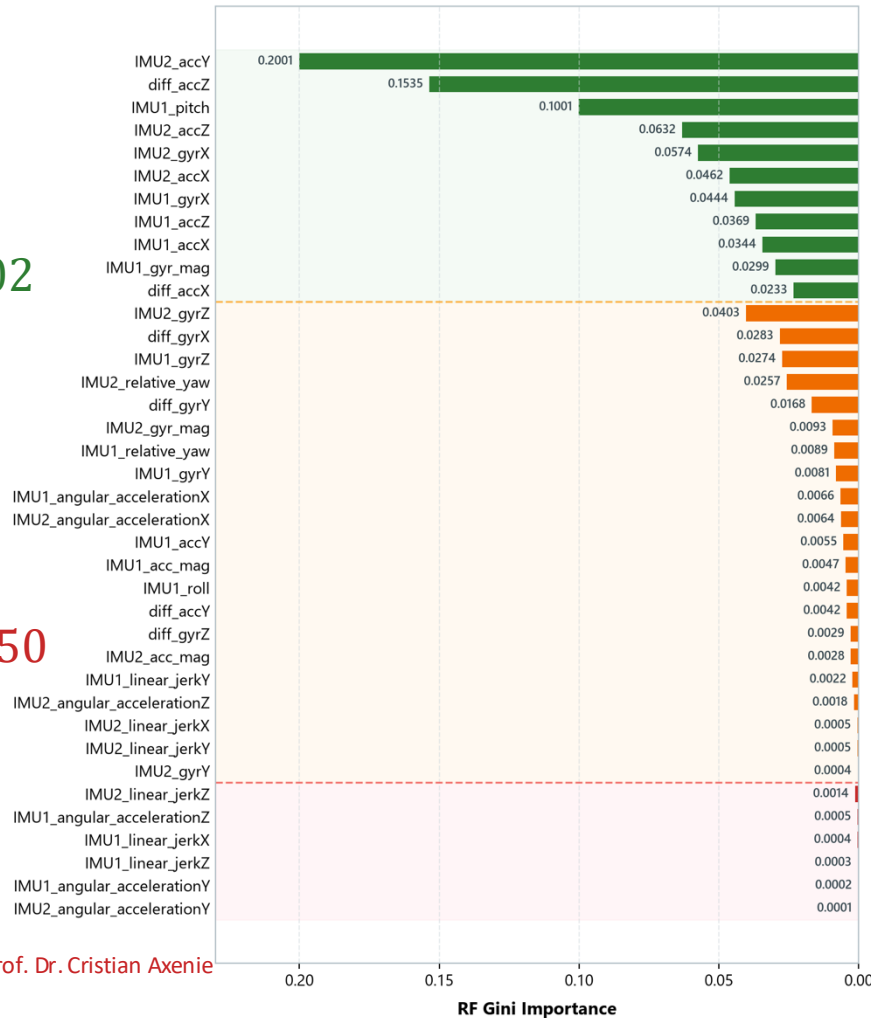


Gesture Recognition With IMU Data

Implementation: Data Audit – Feature Engineering.

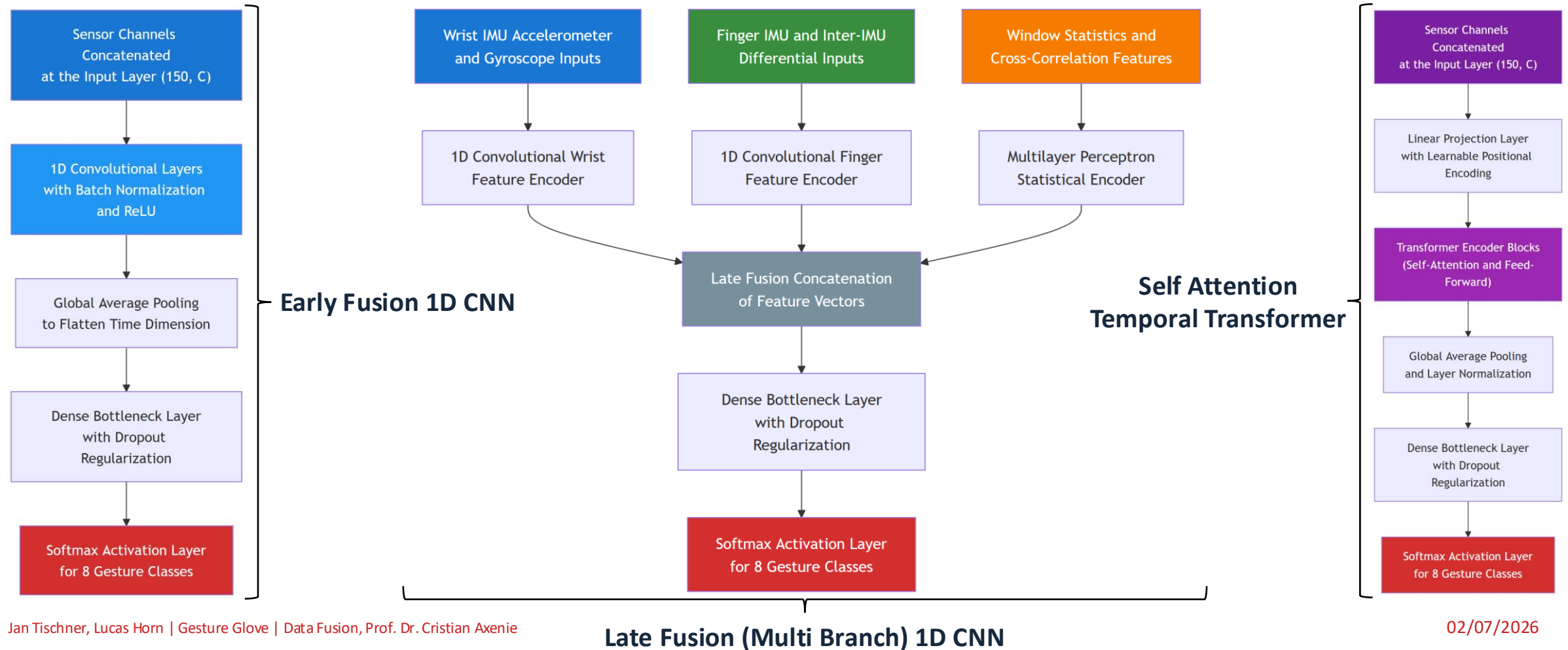
3-tier categorization:

- **Mandatory:**
 $MI > 0.90$ & $RF > 0.02$
- **Dynamic selection.**
- **Pruned:**
 $RF < 0.002$ & $MI < 0.50$



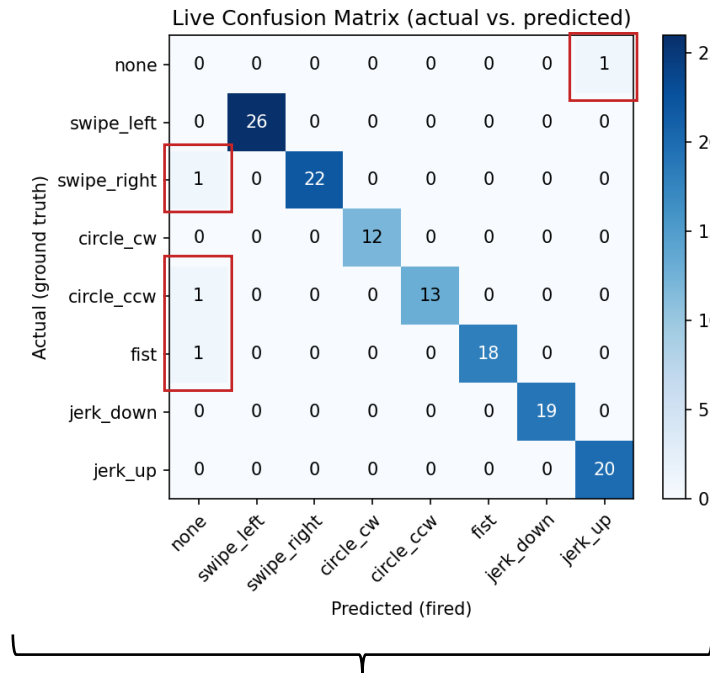
Gesture Recognition With IMU Data

Implementation: Model Architectures.

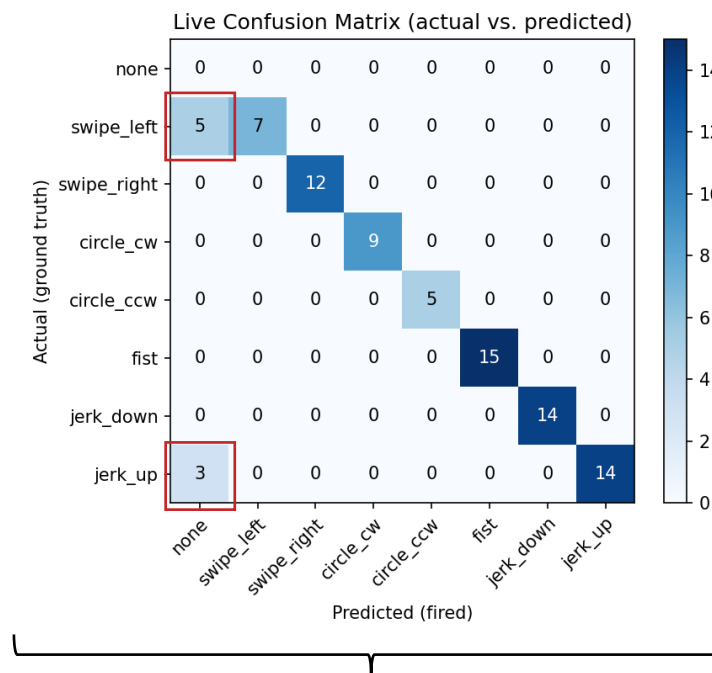


Gesture Recognition With IMU Data

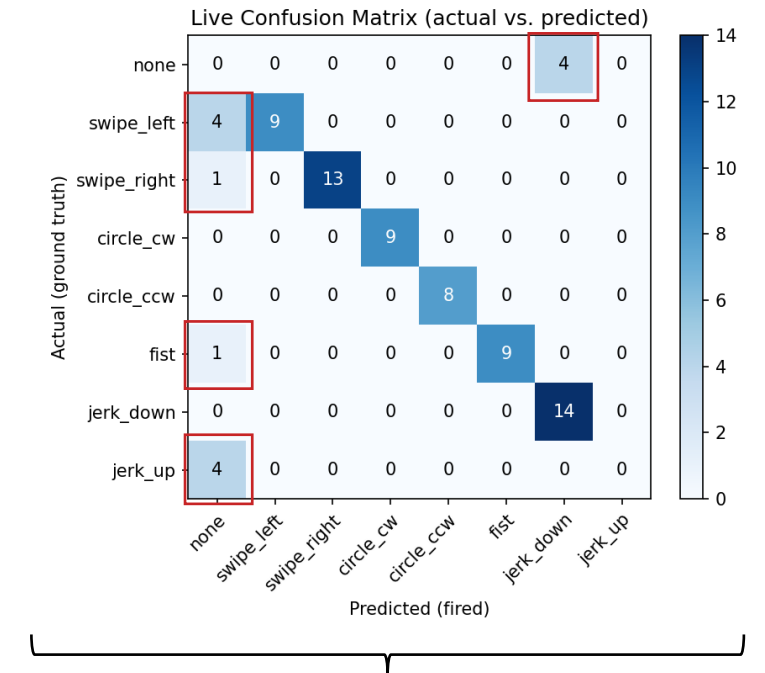
Implementation: Model Architectures – Real World Performance.



Early Fusion 1D CNN



Late Fusion (Multi Branch) 1D CNN



Self Attention Temporal Transformer

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